

The equations of the tangent and the normal

Lessons 17–18 · Algebra 11, §1.6 · P1 · 7.6 · 28 problems

Every answer needs one gradient and one point. The normal gradient is $-\frac{1}{m}$, so a horizontal tangent has a **vertical** normal, which has no gradient at all.

NAME _____

CLASS _____

DATE _____

Easy 7

- | | |
|--|--|
| 1 Tangent to $y = x^2 + 1$ at $x = 2$ | 2 Normal to $y = x^2 + 1$ at $x = 2$ |
| 3 Tangent to $y = x^3$ at $x = 1$ | 4 Gradient of the normal to $y = x^2$ at $x = 3$ |
| 5 Where is the tangent to $y = x^2 + 8x$ horizontal? | 6 Tangent to $y = 5 - 2x$ at $x = 7$ |
| 7 Normal to $y = x^3$ at $x = 0$ | |

Medium 7

- | | |
|---|--|
| 8 Tangent to $y = x^3 - 3x + 1$ at $x = 2$ | 9 Normal to $y = x^3 - 3x + 1$ at $x = 2$ |
| 10 Tangent to $y = \sqrt{x}$ at $x = 9$ | 11 Tangent to $y = \frac{6}{x}$ at $x = 3$ |
| 12 Where has $y = x^2 - 2x$ a tangent parallel to $y = 4x$? | 13 Where has $y = x^3 - 9x$ a tangent parallel to the x -axis? |
| 14 The tangent to $y = x^2$ at $x = 4$ crosses the x -axis where? | |

Hard 7

- 15 The tangent to $y = x^2 - 3x$ at $x = a$ passes through $(0, -4)$. Find a .
- 16 The normal to $y = x^2$ at $x = 2$ meets the curve again. Find the second point.
- 17 Find k so that $y = 2x + k$ is a tangent to $y = x^2 + 3x + 4$.
- 18 Find the tangent and the normal to $y = x^3 - x$ at $x = -1$.
- 19 Show that the tangent to $y = \sqrt{x}$ at $x = a$ cuts the y -axis at $(0, \frac{\sqrt{a}}{2})$.
- 20 Find the two tangents to $y = x^2$ that pass through $(1, -3)$.
- 21 Where do the normals to $y = x^2$ at $x = 1$ and $x = -1$ meet?

Very hard 7

- 22 The tangent to $y = \frac{1}{x}$ at $x = a$ ($a > 0$) meets the axes at P and Q . Show that the point of contact is the midpoint of PQ .
- 23 Find the angle between $y = x^2$ and $y = x^3$ at $(1, 1)$.

24 Find the normal to $y = x^2 - 4x$ at the point where the curve crosses the positive x -axis.

25 Show that every normal to $y = x^2$ except the y -axis meets the curve twice.

26 Find the points on $y = x^3 - 3x^2 + 2$ where the tangent is parallel to $y = 9x$.

27 The tangent to $y = x^2 + bx + c$ at $x = 1$ is $y = 3x - 2$. Find b and c .

28 Find the shortest distance from the origin to the tangent to $y = x^2$ at $x = 1$.
